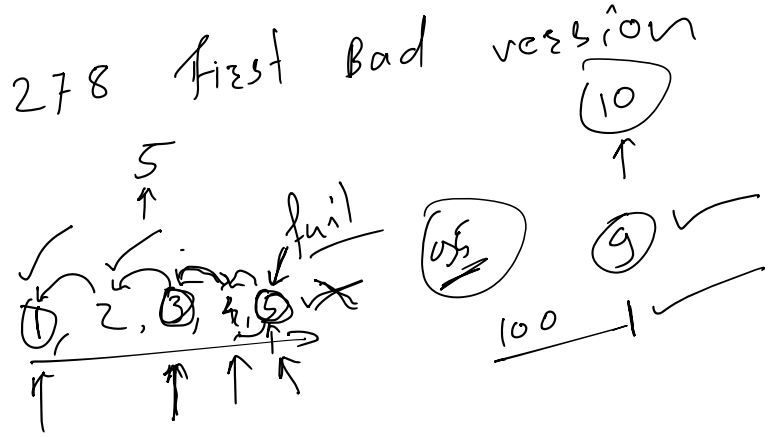
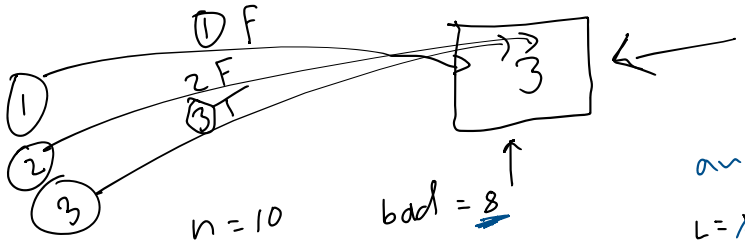
Lec28

13 March 2026 12:26 PM

278 First Bad version





n = 10 bad = 8

1 10
L H

ans = 8

L = ~~1~~ ~~6~~ ~~7~~ 8

H = ~~10~~ 7

mid = ~~5~~ ~~8~~ ~~6~~ 7

7

$\frac{1+10}{2} = 5$

$\frac{6+10}{2} = 8$

$\frac{6+7}{2} = 6$

$\frac{7+7}{2} = 7$

```
while(L<=H ){
    mid = (L+H)/2
    if(isBadVersion(mid) ==True){
        ans = mid
        H = mid-1
    }else{
        L = mid+1
    }
}
```

2B

Int ←

2126753390

$2^{31}-1$

$2^{31}-7$ × $2^{31}-1$

L H

$2^{31}-4$

bad

4B

-3^4

-3^2

31 31

$2^{31} \neq 2^{31}$

2^{32}

2^4

2^4

2×2^3

$mid = lo + (hi - lo)/2$

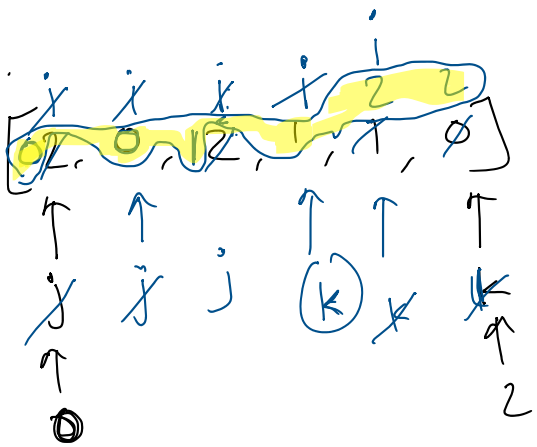
75 sort colors

→ nums = [0, 0, 1, 1, 2, 0]

o/p = [0, 0, 1, 1, 2, 2]

→

0	1	2
red	w	b



i = 0 = [0] = 2

i = 0 = [0] = 0

i = 1 = [1] = 0

i = 2 = [2] = 2

i = 2 = [2] = 1

i = 3 = [3] = 1

i = 0

```
i = 0
j = 0
k = n-1

while(i<=k){
    if(nums[i] == 0){
        temp = nums[i]
        nums[i] = nums[j]
        nums[j] = temp
        j++
        i++
    }
    else if (nums[i] == 2){
        temp = nums[i]
        nums[i] = nums[k]
        nums[k] = temp
        k--
    }
    else {
        i++
    }
}
```